Clinical Risk Assessment Report: Patient 59

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 12.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.55x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.9%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.3 [Bottom <1%] (+1.3%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.69 [Top 17%] (+1.1%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.05 [Top 15%] (+1.0%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.65 [Avg (66th %ile)] (+0.9%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.41 [Avg (29th %ile)] (-3.9%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.21 [Avg (49th %ile)] (-2.0%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

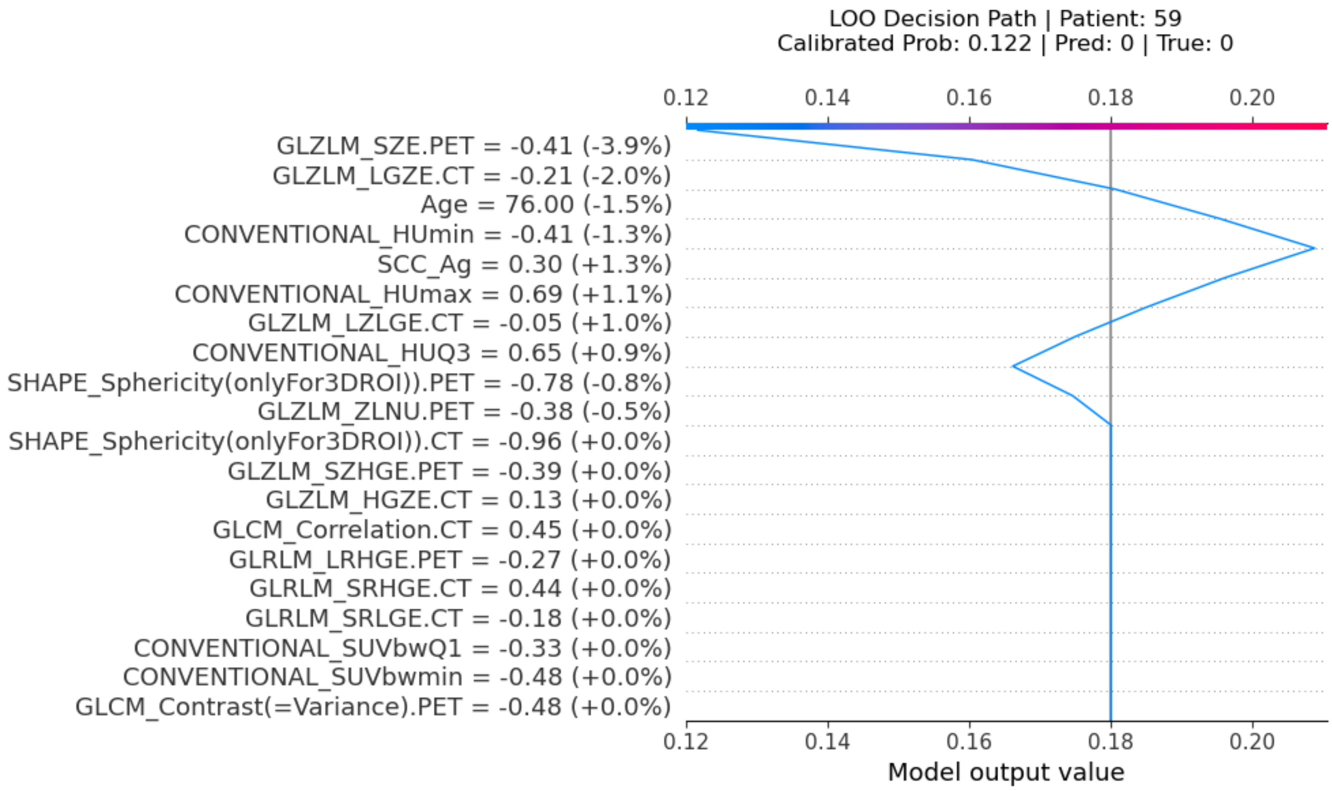
Patient Age (Age) **Value: 76.0 [Top 8%] (-1.5%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.41 [Avg (32th %ile)] (-1.3%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.